

CMSE 801 - Midterm Study Guide

The following document is intended to provide you with a list of the types of tasks you may be asked to complete on your midterm exam for CMSE 801. The exam will be “open internet” so that you can look up python syntax and package documentation. However, you will **not** be allowed to access your email or any other means of person-to-person communication (e.g. Slack) or collaborative tools (e.g. Google Docs). You will also be expected to complete the exam **without** the use of previous assignment notebooks, but maybe bring along handwritten notes if you wish.

For the midterm, you should be comfortable with:

- Using and indexing NumPy arrays (both 1D and 2D arrays)
- Making plots with Matplotlib
- Loading and analyzing data using NumPy
- Writing loops (for and while loops)
- Using `if/elif/else` statements
- Masking NumPy arrays
- Using `odeint` to solve a single differential equation or a system of multiple equations
- Explaining what a compartmental model is, discussing how you would define compartments and connections between the compartments, and describing how you might write a program to evolve a compartmental model

Some or all of these concepts may show up on the exam, so make sure you spend time reviewing the ones you’re less comfortable with and don’t hesitate to reach out on Slack to ask your fellow classmates or the instructors questions about things you don’t understand.

Final Note: If you’re currently having any issues with your Python installation or any other software related to the course, contact your instructor **as soon as possible** and explain the issue so that it can be addressed before the exam. Also, make sure that you can successfully log into `jupyterhub.msu.edu` so that you can use it as a backup if something goes wrong on exam day.